5-TIER CYBER-PHYSICAL SYSTEM ARCHITECTURE DIAGRAM

AI-Powered Smart Campus Resource Management System | Course: E1CSA313 | Batch: DP1-24-28-CSE-GEN-L-01

ALLIANCE UNIVERSITY
ASAC CSE · SEMESTER V

TIER 5: HUMAN-MACHINE INTERACTION & ACTUATION DOCK (CLIENT)

Zero-install responsive frontend delivering multi-role views, live spatial telemetry & sub-second override controls

- Central Multi-Role Dashboard
- Interactive Building Drawer
- Dynamic D3 Sankey Power Flow
- **Manual Actuator Override Dock**

TIER 4: 3D WEBGL GRAPHICS & SPATIAL SHADER ENGINE

Procedural Blender geometry pipeline rendering campus assets at 60 FPS under 15 MB VRAM footprint

- Three.js r170 Scene Graph
- OrbitControls & Frustum Culling
- Dynamic Thermal Shaders
- Low-Poly Parametric GLB <15MB

TIER 3: COMPUTATIONAL INTELLIGENCE & DECISION ENGINE

Core algorithmic layer executing space reclamation, microgrid peak-shaving dispatch & timetable constraint solving

- **Timetable CSP Solver** — Eliminates 100% room clashes
- **15-Min Hysteresis Buffer** — Zero false-alarm chiller alerts
- **1.2 MWh BESS Dispatch** — ToD peak tariff shaving (>18%)
- **Thermal Euler Differential Solver** — Lumped-capacitance building dynamics

TIER 2: IOT TELEMETRY & REAL-TIME DATA INGESTION ENGINE

Physics-coupled state engine synthesizing 1,000+ virtual sensor beacons, ERP schedules & weather inputs

- 1,000+ Spatial Sensor Beacons
- Alliance ERP Timetable Scraper
- BESCOM Tariff & Meter Feeds
- WebSocket State Hydration <100ms

TIER 1: PHYSICAL CAMPUS INFRASTRUCTURE & ELECTRICAL ASSETS

Alliance University 60-acre Anekal campus buildings, renewable solar fields & power distribution equipment

- Admin & Library Blocks
- ACED Labs & Classrooms
- 480 kWp Rooftop Solar PV
- 1.2 MWh LiFePO4 BESS
- Central Chiller Plant

DRAWING NO: AU-DP1-DWG-001 | SHEET 1 OF 5 | SCALE: N.T.S. | PROJECTION: THIRD ANGLE

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